

Coarse latent states with learned subpatch moments for multi-resolution ocean emulation

Final status (2026-07-26): S1 reconstruction, S2 objective selection, and the promoted full one-/half-degree S3 training, independent validation, checkpoint-state audit, and 65-batch data audit are complete. No quarter-degree result is included.

Abstract

The `SamudraMulti` architecture on `main` maps every physical resolution to a fixed 60×72 processor grid, but its Perceiver encoder compresses each fixed $3^\circ \times 5^\circ$ patch without an explicit resolved-value route and its Perceiver IO decoder must learn spatial routing and channel rendering jointly. Earlier controlled experiments showed that this decoder dominates the unexpected error on the nominally trivial auto-encoding task. A native-grid learned inverse fixes reconstruction but no longer tests the desired scientific regime: a processor state spatially coarser than the observations.

We therefore retain the 60×72 processor grid and replace the representation heads with (i) a cosine-latitude-weighted resolved mean plus 16 learned continuous within-patch moments and (ii) a coordinate-resampling base plus a zero-initialized, position-anchored local attention residual. The inverse is trained jointly, then frozen. A shared processor advances the latent state once per physical time step; a separate boundary encoder supplies exactly one aligned forcing state to each invocation. At matched one-/half-degree reconstruction budget, the learned decoder reduces wet-cell MSE by 62–72% relative to a bilinear-only decoder and retains 13–22 times more target gradient power. In a four-objective dynamics screen, physical loss plus a 0.1-weighted stop-gradient latent-teacher loss wins all aggregate leads and 11 of 12 exact route/lead comparisons. Removing the learned subpatch channels after training increases lead-one raw physical MSE by factors of 2.4–2.9, demonstrating that the S2 forecast depends on information beyond patch means through the frozen decoder, processor, or both. The remaining concerns are half-degree lead-one skill and weak velocity high-wavenumber power. The S3 training procedure runs 6,336 updates—within 0.9% of the 6,392-update native reference—and selects its epoch-12 checkpoint after 4,224 updates. That checkpoint reaches aggregate lead-one, lead-two, and lead-four normalized MSE of 0.0828/0.1008/0.1281 and beats persistence on all 12 route/lead comparisons. This establishes useful coarse-latent dynamics, but errors remain 1.9–2.1 times those of the native-grid model and half-degree velocity spectra remain severely attenuated. We therefore retain the new inverse as the preferred coarse-latent research architecture, but not as a replacement for the native-grid production baseline. The combined audits isolate two remaining limitations: the frozen inverse already loses substantial fine-scale power at depth zero, and teacher-latent error grows with lead even as the two input resolutions converge toward a common latent trajectory.

For reproducibility, “`main`” denotes `origin/main` commit `d689f92c`, whose `SamudraMulti` and `default model configuration` were inspected on 2026-07-24.

Architecture and experimental contract

Property	SamudraMulti on main	Coarse moment/attention model
Processor grid	Fixed 60×72	Fixed 60×72
Patch encoder	Perceiver compression to one token	Explicit cosine-latitude mean plus learned subpatch moments
Decoder	Perceiver IO learns routing and channels jointly	Coordinate base plus anchored local attention residual
Query role	Position query primarily controls attention	Position anchors logits, conditions values, and enters hidden output directly
Geometry	Added to reconstructive representation	Separate learned processor sidecar
Boundary forcing	Mixed with state before encoding	One separately encoded boundary state per processor step
Autoregression	Decode and re-encode between steps	Encode once and remain latent
Inverse during dynamics	Jointly trainable	Frozen and bitwise audited
Output resolution	Flexible query grid	Flexible coordinate grid

For patch p , physical cell i , channel vector x_{pi} , and cosine-latitude weight $a_{pi} = \cos(\varphi_{pi})$ evaluated at the cell center, the encoder first computes

$$\mu_p = \frac{\sum_i a_{pi} x_{pi}}{\sum_i a_{pi}}.$$

A coordinate network learns 16 scalar basis functions $\phi_m(r_{pi})$ of evenly spaced, index-normalized within-patch coordinates. Each basis is cosine-latitude-weight centered and RMS-normalized within the actual input-resolution patch, giving

$$q_{pcm} = \frac{\sum_i a_{pi} (x_{pic} - \mu_{pc}) \tilde{\phi}_m(r_{pi})}{\sum_i a_{pi}}.$$

The 160-channel token is

$$z_p = [W_\mu \mu_p, W_q \text{vec}(q_p)],$$

with 40 resolved-mean channels and 120 projected moment channels. These are learned coordinate moments, not 16 fixed statistical moments. The same coordinate network operates on 3×5 one-degree patches and 6×10 half-degree patches, so both inputs produce 60×72 tokens.

The decoder is

$$D(z; \ell_o) = B(z; \ell_o) + C(z; \ell_o).$$

B is periodic-longitude physical-coordinate bilinear resampling followed by a shared output-channel projection. C attends over a 3×3 neighborhood of coarse tokens. Its attention logits include a strong distance-to-query anchor; its values receive the unnormalized token and the continuous query-to-token offset; and a direct query residual carries absolute spherical position and source/output scale into the decoded hidden state. Only the latent-content input to attention keys is normalized; the correction feed-forward block has its own hidden-state LayerNorm. The value route, base route, and explicit resolved means retain an unnormalized physical-amplitude path. The correction output is initialized to zero, so optimization begins from B , but the converged correction is not constrained to remain small.

After inverse training, encoder and decoder parameters are frozen and one physical step is

$$z_m = z_{m-1} + \alpha \odot P(z_{m-1} + E_b(b_m) + G(\ell_z)).$$

P is the shared nonlinear processor, E_b is a separate boundary encoder, and G is a learned, zero-initialized projection of deterministic spherical position and cell-area features. The learned $\alpha \in \mathbb{R}^{1 \times 160 \times 1 \times 1}$ scales processor residual channels; it is unrelated to the decoder correction. A lead N means N autoregressive processor applications and N boundary-encoder calls, advancing the forecast by $N dt$. Intermediate states remain latent: they are not decoded and re-encoded.

The selected dynamics objective is

$$L = L_x(D(z_N), x_{t+N}) + 0.1 \frac{\sum_{b,c,p} m_{bp} (z_{N,bcp} - \text{sg } E(x_{t+N})_{bcp})^2}{\sum_{b,c,p} m_{bp}},$$

where m marks wet coarse tokens and sg stops gradients through the target encoding.

Evidence

The S1 inverse was tested with two independent learned-decoder seeds and a matched bilinear-only control. Across all one-/half-degree input/output routes, the learned decoder gives wet-cell MSE 0.097–0.183, versus 0.353–0.482 for bilinear. It retains target gradient-power ratios 0.39–0.70, versus 0.019–0.056, without a patch-seam error spike. The two learned seeds agree closely. Bilinear produces higher synchronized-resolution latent cosine (0.974 versus 0.961) while losing most fine structure, showing why latent agreement alone is not a suitable promotion criterion.

S2 froze the selected seed-15 inverse and trained four matched processor objectives for 768 updates on the half-degree-only proxy. Subsequent full held-out-year transfer validation over all four input/output routes gives:

Objective (w_x, λ_z)	Lead 1	Lead 2	Lead 4	Persistence reduction
Physical only (1, 0)	0.1038	0.1322	0.1653	3.3% / 25.5% / 34.6%
Latent only (0, 1)	0.1042	0.1313	0.1620	2.9% / 26.1% / 35.8%
Combined (1, 0.01)	0.1029	0.1308	0.1636	4.2% / 26.3% / 35.2%
Combined (1, 0.1)	0.1013	0.1283	0.1608	5.7% / 27.7% / 36.3%

The promoted combined objective is best in 11/12 exact route/lead cells. The exception is one-degree to one-degree at lead four, where latent-only is 1.4% lower. The promoted model beats persistence at every lead-two and lead-four route, but remains 3.7% worse at half-degree same-grid lead one. Aggregate high-wavenumber power ratios are 0.791/0.916/0.336/0.449/0.655 for $\theta_{\text{tao}}/\text{so}/\text{uo}/\text{vo}/\text{zos}$: scalar fields are credible, but velocity is still too smooth.

Checkpoint audits verify that all 30 inverse tensors remain bit-identical. Zeroing the 120 moment channels while retaining the 40 mean channels increases lead-one raw MSE from 0.136 to 0.391 on the one-degree grid and from 0.221 to 0.519 on the half-degree grid. Same-latent cross-output patch-mean symmetric normalized MSE stays near 0.017 through lead four. Zero boundary forcing increases aggregate lead-four error by 4.6%, and reversed forcing order by 1.0%; the boundary path is causal but not yet strongly used in this short screen.

The principal forecast comparators have deliberately different scopes:

Model	Processor grid	Training scope	Lead 1	Lead 2	Lead 4
main Perceiver model	60×72	full one-degree, historical contract	0.29469†	—	—
Native-grid learned inverse	input grid	full one-/half-degree, 6,392 updates	0.03982	0.05408	0.06595
Coarse moment/attention S2	60×72	768-update objective screen	0.10125	0.12831	0.16083
Coarse moment/attention S3	60×72	6,336-update procedure; selected at 4,224	0.08275	0.10085	0.12806

†The `main` value is its one-degree same-grid lead-one result, not an aggregate over four routes. It is useful historical context but not a controlled single-factor comparison. The native and S3 runs share the one-/half-degree data interval, route schedule, true processor depths, frozen-inverse temporal contract, approximately matched procedure-level update ceiling, and validation year. They differ in latent grid, objective (physical-only native versus physical plus 0.1 latent teacher), global batch (30 versus 32), and selected checkpoint update. This is an architecture/training-package comparison, not a controlled latent-grid-only ablation.

The independent selected-checkpoint S3 validation gives:

Route	Lead 1	Lead 2	Lead 4	Persistence reduction
$180 \times 360 \rightarrow 180 \times 360$	0.06064	0.07776	0.10305	27.6% / 48.6% / 54.0%
$180 \times 360 \rightarrow 360 \times 720$	0.10411	0.12366	0.15324	25.1% / 35.2% / 38.5%
$360 \times 720 \rightarrow 180 \times 360$	0.06245	0.07859	0.10310	29.5% / 49.6% / 55.0%
$360 \times 720 \rightarrow 360 \times 720$	0.10381	0.12339	0.15286	12.2% / 41.9% / 50.3%

Zeroing the per-step boundary states raises aggregate error by 7.7%/13.7%/18.9% at leads one/two/four, while reversing their order raises it by 3.1%/3.1%/4.7%. Thus one boundary encoding per processor invocation is not merely present in the graph: the forecast uses its magnitude and temporal alignment increasingly with lead.

Full training improves selected-checkpoint high-wavenumber power ratios to 0.830/0.937/0.380/0.523/0.739 for `thetao/so/uo/vo/zos`. The scalar improvement is credible, but velocity remains the decisive failure. On half-degree outputs, `uo/vo` ratios are only 0.323/0.142 for a one-degree input and 0.295/0.130 for a half-degree input. These losses coexist with good persistence skill and stable patch-seam ratios, so aggregate MSE alone would overstate the scientific closure.

Direct checkpoint inspection selects epoch 12 (4,224 updates, `best_val_loss=0.0827667`) and verifies all 30 frozen inverse tensors/536,436 parameters are bit-exact to S1. The terminal epoch-18 checkpoint records 6,336 updates, 56 short of the configured target because the epoch bound fired first. Its late lead-four degradation is therefore real but not the promoted endpoint. The final data audit covers all 65 synchronized validation batches. One- versus half-degree latent cosine improves from 0.961 before processing to 0.978 at lead four, and cross-output patch-mean symmetric normalized MSE remains 0.0173–0.0174 at every lead. However, forecast-to-teacher latent symmetric normalized MSE grows from 0.097/0.109 at lead one to 0.257/0.278 at lead four on one-/half-degree grids. The processor moves the two resolutions toward a common trajectory while becoming increasingly inconsistent with the corresponding encoded future states.

Zeroing initial moment channels raises raw forecast MSE by factors of 3.55/2.78/2.14 on the one-degree same-grid path and 2.69/2.27/1.85 on the half-degree same-grid path for leads one/two/four. This proves forecast dependence on moment channels for the trained model, but does not separate their direct frozen-decoder contribution from their processor contribution. The declining ratio with lead is consistent with progressive loss of initial subpatch information, although it is not sufficient alone to establish that mechanism. Zero-boundary latent differences grow

approximately fourfold from lead one to lead four, independently confirming that forcing magnitude affects repeated transitions. The separate time-reversal physical-loss ablation supports temporal alignment of that forcing.

Discussion and recommendation

The evidence rejects two simple explanations of the original decoder failure. It is not primarily a shortage of Perceiver decoder latents: increasing them did not resolve the controlled auto-encoding error. It is also not evidence that a coarse latent is intrinsically unusable. Rather, the evidence supports an explicit low-frequency route, learned phase-sensitive subpatch summaries, and a decoder whose output query is spatially anchored and directly represented in the value/rendering path. The selected decoder is therefore a form of position-anchored direct cross-attention, used as a learned residual around a stable coordinate route.

The S2 result also argues against adding an explicit cross-resolution latent-agreement penalty now. Physical-only training produces the strongest cross-resolution latent agreement at lead four but worse forecasts, while the combined objective improves physical skill and teacher-latent accuracy without collapsing resolution-specific subpatch state. Likewise, the forecast-dependence intervention, together with the synthetic mean-only control, argues against reverting immediately to a mean-only or purely linear restriction; it does not rule out a separately optimized mean-only architecture.

S3 resolves the conditional recommendation. Retain the patch-moment encoder, continuous position-anchored hybrid decoder, per-step boundary encoder, and 0.1-weighted latent teacher as the current coarse-latent architecture. They support zero-to- N processor applications, flexible output grids, causal boundary use, and useful four-route skill. Do not replace the completed native-grid model: the selected coarse checkpoint has 2.08, 1.86, and 1.94 times its lead-one/two/four MSE. Because it was selected earlier and used a different objective and global batch, those ratios compare the promoted training packages rather than isolate spatial latent resolution.

The next falsifiable hypothesis should target the unresolved representation bottleneck rather than revert to Perceiver IO or bilinear decoding. Test a split subpatch state with variable-grouped coefficient blocks—at minimum scalar and horizontal-velocity blocks—whose continuous basis functions are encoded on the physical cells, evolved on the same 60×72 grid, and evaluated directly at requested output coordinates. Keep the present anchored attention as a residual for cross-patch consistency. This preserves flexible resolution while testing whether dedicated velocity phase/orientation capacity improves on the shared 16-basis, 120-channel projection. Processor smoothing, loss weighting, and the frozen decoder remain alternative causes because the final S3 audit does not isolate those components individually. Pair the inverse test with variable-wise gradient/spectral loss and require velocity high-wavenumber closure before another full dynamics run.

After that inverse-side gate passes, reopen the processor as a separate multi-step latent-transition problem. The final audit shows that resolution agreement improves while teacher-latent error grows, so test explicit coupled updates for resolved-mean and subpatch-coefficient blocks and select on multi-lead teacher error as well as decoded physical loss. The full S3 training package improves S2 forecast losses by roughly 18%–21%, but route exposure, global batch, initialization, and schedule also changed. The persistent spectral deficit and late-training lead-four drift do not support width alone as the first remedy. Use multi-lead checkpoint selection and report both best and terminal weights. Do not start quarter-degree validation before reviewing this one-/half-degree endpoint.

Glossary and implementation map

- **Fixed coarse processor grid.** `patch_extent: [3.0, 5.0]` makes the encoder aggregate 3×5 one-degree cells or 6×10 half-degree cells into each token. Both inputs therefore produce tensors of shape `[B,160,60,72]`; the processor never operates on the requested output grid. The exact patch construction and divisibility checks are in [PatchMomentEncoder.output_resolution](#).
- **PatchMomentEncoder / learned moments.** The exact equations above are implemented in [encoder.py](#). Configuration: `encoder.patch_moment_count: 16`, `encoder.patch_mean_channels: 40`, and `patch_extent: [3.0, 5.0]` in [model_iterable_inverse_coarse_moment_attention.yaml](#).
- **Frozen inverse.** The jointly trained mapping $D \circ E$ from physical state to coarse latent and back. “Frozen” means every `encoder.*` and `decoder.*` tensor is excluded from S2/S3 optimization and checked bitwise against the S1 checkpoint. Configuration: `frozen_model_prefixes: ["encoder.", "decoder."]` in [train_cross_1_halfdeg_coarse_latent_dynamics_full.yaml](#).
- **Continuous anchored hybrid decoder.** $D = B + C$, implemented by [ContinuousResampleAttentionResidualDecoder](#). Configuration begins with `decoder.continuous_resample_attention_residual: true`; radius 1 means nine candidate coarse tokens, four heads of width 32, and position-bias strength 8.0.
- **Zero-initialized attention residual.** Only C ’s final output projection starts at zero. It is free to become large; S1 correction RMS is 0.79–0.81 of prediction RMS. This is distinct from processor residual scale α .
- **Position-anchored direct cross-attention.** The decoder correction adds $-8\|r_q - r_k\|^2$ to attention logits, conditions each value on continuous query-to-token offset, and adds continuous absolute query/scale features directly to the hidden output. See [ContinuousCoordinateAttentionCorrection](#).
- **Decoder query and channel semantics.** There is one query per requested output coordinate, not one query per target channel. Each query produces the entire prognostic-channel vector, so the selected decoder maps `[B,160,60,72]` to `[B,C_out,H_out,W_out]`; no target-channel identity is embedded in the query. The final [output_projection](#) performs this hidden-to- C_{out} mapping.
- **LayerNorm concern.** `content_norm` normalizes only attention keys. Values consume the unnormalized latent token; the bilinear base and resolved-mean route are also unnormalized. This keeps a direct amplitude-bearing path while allowing normalized similarity matching.
- **Geometry sidecar.** `encoder.geometry_mode: sidecar` means absolute position and grid scale are not added to reconstructive encoder content. Instead, [ProcessorGeometryConditioner](#) supplies them to each processor call.
- **Separate boundary encoder.** `boundary_encoder` maps exactly the forcing state b_m aligned with transition m onto the latent grid. It is called once per processor application; see [SamudraMulti.process](#) and [SamudraMulti.latent_rollout](#).
- **Latent autoregression.** Encode once, apply the processor N times, and decode selected latent states. A lead N is not one processor call with N boundary states. Code: [SamudraMulti.latent_rollout](#).
- **True-depth sampling.** `train_processor_depths: [1,2,4]` selects one true physical lead per training batch. Depth N consumes labels and ordered boundary states through $t + N dt$; it is not an internal repeated block at a fixed target time. Validation always reports all three depths. Configuration: [train_cross_1_halfdeg_coarse_latent_dynamics_full.yaml](#).
- **Processor residual scale α .** A learned tensor of shape `[1,160,1,1]` used in $z + \alpha \odot P(\cdot)$. It is initialized to zero by `processor_residual: true`. In the selected S3 checkpoint its mean absolute

value is 0.1019; 0.0901 over the first 40 resolved-mean channels and 0.1058 over the 120 subpatch-moment channels. Its sign is not a mixture fraction, because the learned processor output can absorb an arbitrary sign. Exact values and checkpoint comparison are emitted by [audit_checkpoint_state.py](#).

- **Physical and latent objectives.** `physical_forecast_loss_weight` is w_x ; `latent_teacher_loss_weight` is λ_z . The target latent is a no-gradient encoding of x_{t+N} , masked at wet coarse tokens. See [SamudraMulti.forward](#) and [SamudraMulti.latent_teacher_loss](#).
- **Mean-only-initial ablation.** An inference-time causal intervention, not a separately optimized OM4 model: zero the 120 moment channels of the initial latent, retain the 40 mean channels, then run the same trained processor and boundary sequence. It tests forecast dependence on initial moment channels; it does not distinguish their direct frozen-decoder contribution from their processor contribution. Implementation: [audit_coarse_dynamics.py](#).
- **Flexible output resolution.** The decoder accepts requested latitude and longitude arrays and evaluates $B + C$ at those coordinates. Flexibility does not imply that arbitrary fine detail can be recovered from a coarse input.
- **Proposed variable-grouped subpatch coefficients (not implemented).** Replace the single shared subpatch projection with blocks $s_p = [s_p^{\text{scalar}}, s_p^u, s_p^v]$, each computed from continuous within-patch basis moments and evolved on the same coarse grid. The decoder evaluates the corresponding basis block at each requested coordinate before applying the existing anchored attention residual. This is a falsifiable next experiment, not an established remedy or a flag in the current configuration.
- **S1/S2/S3.** S1 trains and audits the inverse; S2 freezes it and selects the dynamics objective in a short matched screen; S3 trains a fresh processor/boundary path at the full one-/half-degree update budget. The authoritative ledger is [coarse_latent_highres_dynamics_results.md](#).